آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre SSC-II General Mathematics Annual Examinations 2025

Introduction

This document has been prepared for the teachers and candidates of Secondary School Certificate (SSC) Part II (Class X) General Mathematics. It contains comments on candidates' responses to the 2025 SSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used. It is imperative to refer to command word guide available on AKU-EB website for understanding the expectations of the command word.

General Observations

Candidates performed reasonably well in some concepts, such as, payment in monthly installments, median, arithmetic mean, volume of a right circular cone. However, candidates who did not score well mostly failed to understand the demands of the questions, often misinterpreting the command words and the stimuli.

Mentioned below are few concepts that teachers need to focus so that the candidates may perform better.

- Income Tax
- HCF and LCM

- Square Root by long division
- Quadratic Equation
- Properties of Parallelogram
- Area of Shaded Region

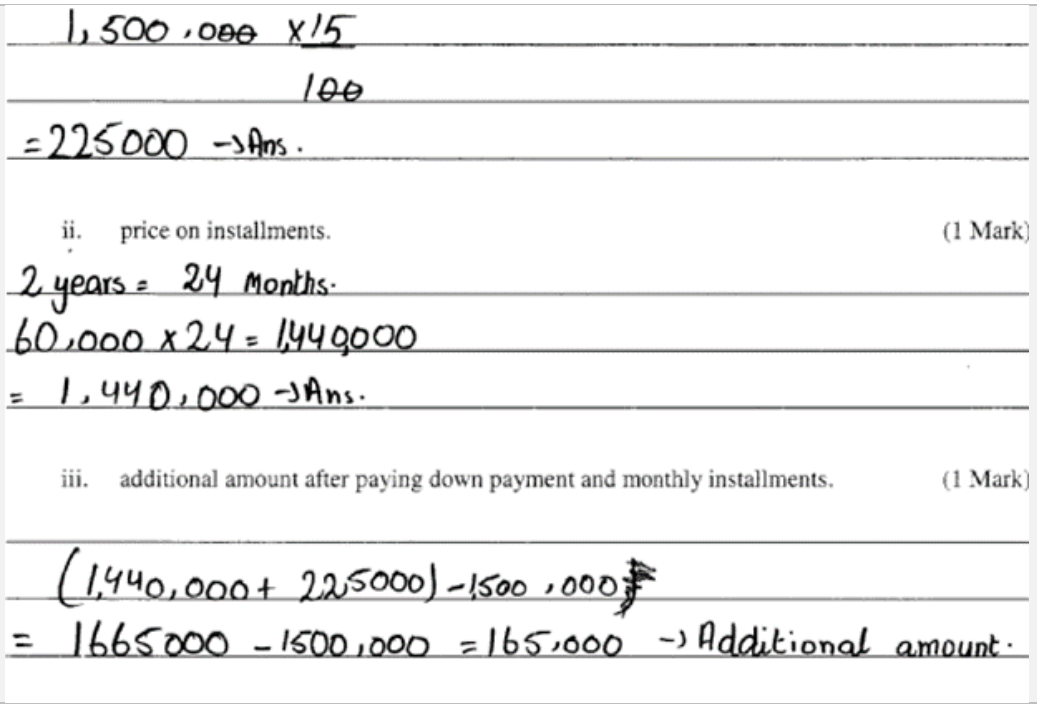
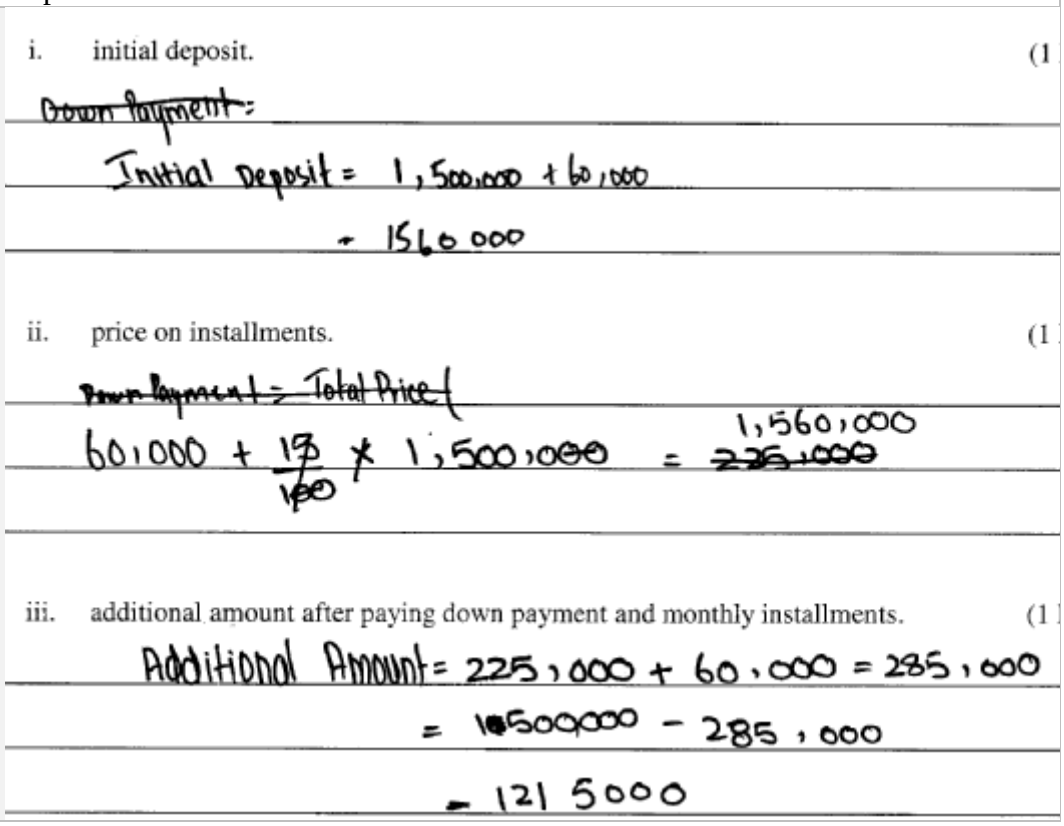
Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS
Constructed Response Questions (CRQs)

Question No. 1a

Candidates were given the choice to attempt any ONE out of the two questions: 1a and 1b. Most of the candidates attempted 1a.

Question Text	A dealer sells a car at cash price of Rs 1500,000. If Asif buys this car on installments, then he would need to pay 15% down payment and the remaining amount in monthly installments of Rs 60,000 for 2 years. In the given situation, calculate the i. initial deposit. ii. price on installments. iii. additional amount after paying down payment and monthly installments.
SLO No.	11.4.2
SLO Text	Solve problems related to SLO 11.4.1. (payment in monthly installments).
Max Marks	3
Cognitive Level	A*
Checking Hints	i. 1 mark for calculation of initial instalment ii. 1 mark for finding total payable amount. iii. 1 mark for calculating the additional payment as a result of installments.
Overall Performance	Few candidates attempted it successfully demonstrated a sound understanding of the practical context involved. They were able to accurately calculate values such as the down payment and monthly instalment. However, candidates who struggled with this question appeared to have difficulty grasping the underlying concept, leading to incorrect evaluation of the required values.
Description of Better Responses	*Better responses* showed all the necessary steps required to solve the given question. In part (i), the candidates accurately computed the initial deposit by taking 15% of the original price. In part (ii), the amount due was calculated correctly $1500{,}000 - 225{,}000 = \text{Rs } 1{,}275{,}000 = 60{,}000 \times 24$. In part (iii), the candidates determined the monthly instalment and accurately evaluated the additional amount paid under the hire purchase system.

Image of Better Response	
Description of Weaker Responses	*Weaker responses* showed conceptual gaps and errors in value interpretation. The candidates calculated an incorrect deposited amount, misapplying the 15% requirement and added 1,500,000 in 60,000. The remaining amount and total payable amount were also calculated inaccurately, reflecting a lack of understanding of the context and requirements of the question. In part (iii), the candidates did not correctly follow the steps needed to determine the required difference, took the difference of 1,500,000 and 285,000 which led to an inaccurate response overall.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

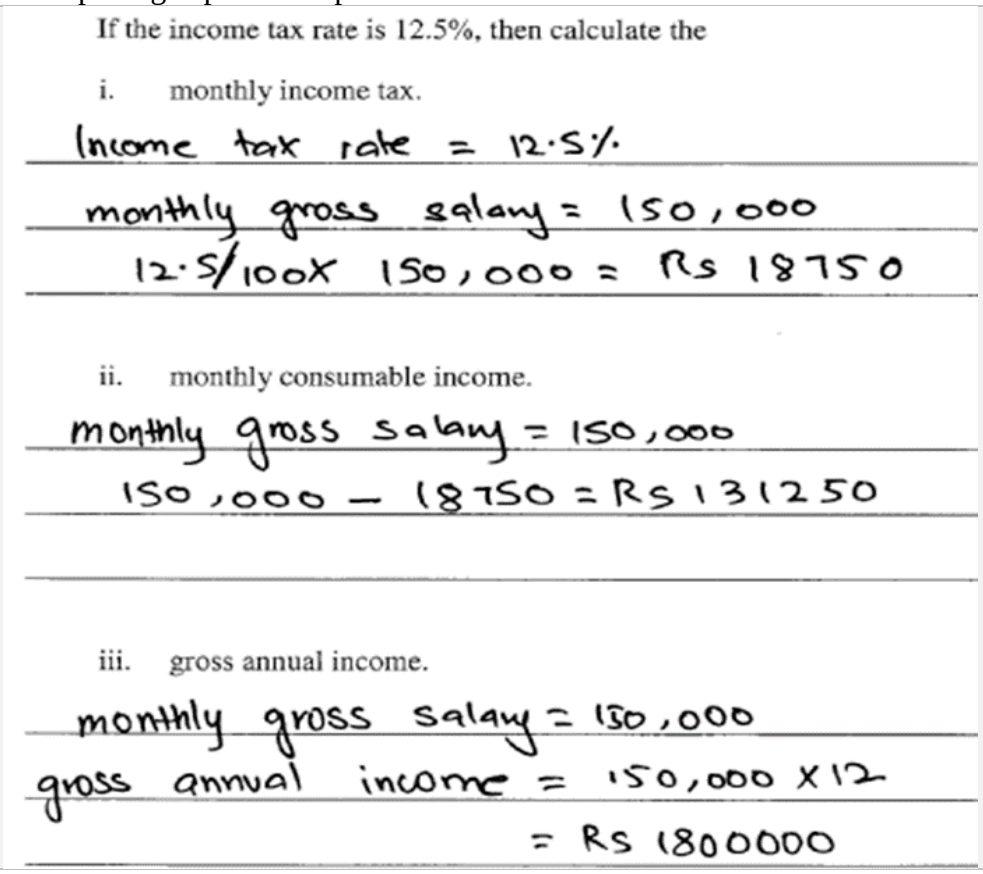
Maximising SLO Achievement	Preferred Pedagogy** Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

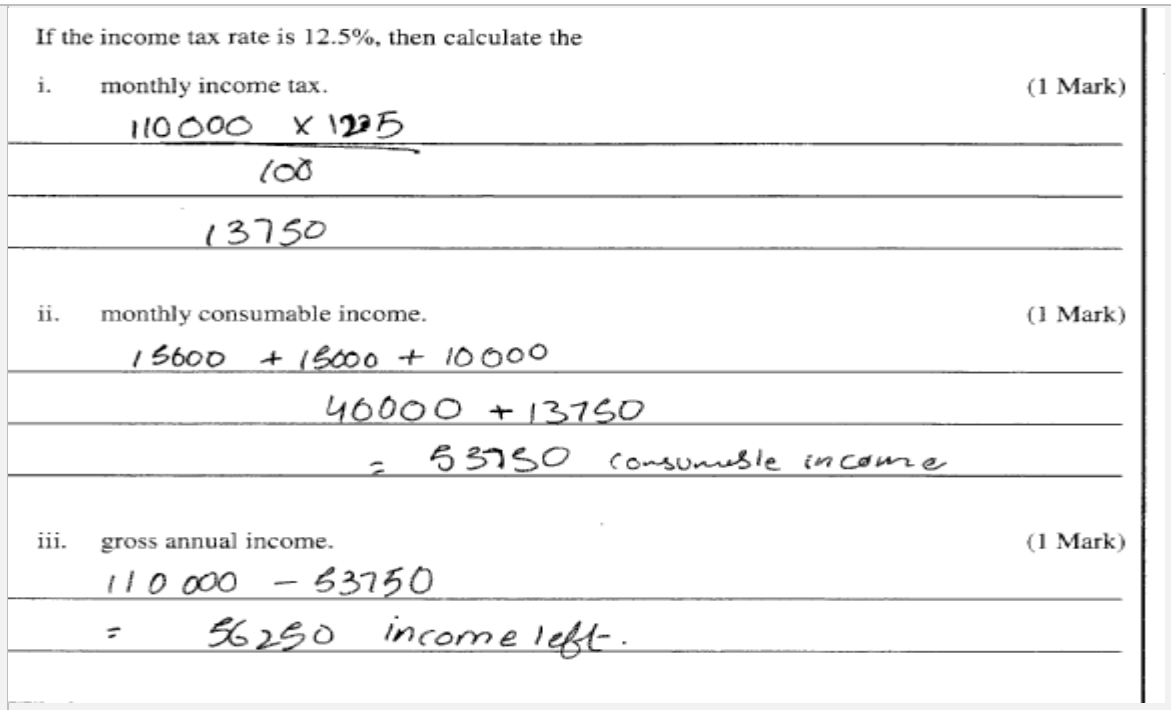
Any Additional Suggestion:
Teachers should provide targeted practice on real-life application problems, such as hire purchase scenarios, ensuring that students understand the step-by-step breakdown of percentage calculations, instalments, and total cost comparisons. Emphasis should be placed on interpreting word problems carefully, identifying relevant values, and clearly distinguishing between one-time and recurring payments.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 1b

Question Text	Sharif earns Rs 150,000 as monthly gross salary. His salary break up is tabulated as given. **Basic Salary (Rs)** / **Medical Allowance (Rs)** / **Food Allowance (Rs)** / **Professional Allowance (Rs)** 110,000 / 15,000 / 15,000 / 10,000 If the income tax rate is 12.5%, then calculate i. monthly income tax. ii. monthly disposable income. iii. gross annual income.
SLO No.	12.3.3
SLO Text	Calculate net income taking into account assorted deductions. (income tax, etc)
Max Marks	3
Cognitive Level	A
Checking Hints	i. 1 mark for calculation of monthly income tax. ii. 1 mark for finding monthly disposable income. iii. 1 mark for calculating the gross annual income.
Overall Performance	Overall, the candidates' performance was below expectations. Those who grasped the concept of income tax and disposable monthly income performed well, demonstrating a

	clear understanding of the question. It was observed that some candidates faced difficulty in calculating the consumer's monthly income.
Description of Better Responses	In *better responses*, the candidates successfully evaluated the correct value of income tax by applying the appropriate percentage $\frac{12.5}{100} \times 150{,}000$. They correctly calculated the monthly consumable income by subtracting the income tax from the gross salary $150{,}000 - 18{,}750$. Furthermore, the candidates demonstrated a clear understanding of annual gross income and accurately multiplied by 12 to arrive at the correct answer. The response reflected a sound conceptual grasp of the topic.
Images of Better Responses	
Description of Weaker Responses	In *weaker responses*, the candidates incorrectly divided the percentage by the given value, which led to a wrong calculation. The monthly consumable income was obtained by adding all values together 15,600 + 15,000 + 10,000, indicating a lack of conceptual understanding. Additionally, the distinction between monthly and annual income was not evident, as the candidates merely added figures without applying the appropriate logic, resulting in an incorrect final answer.

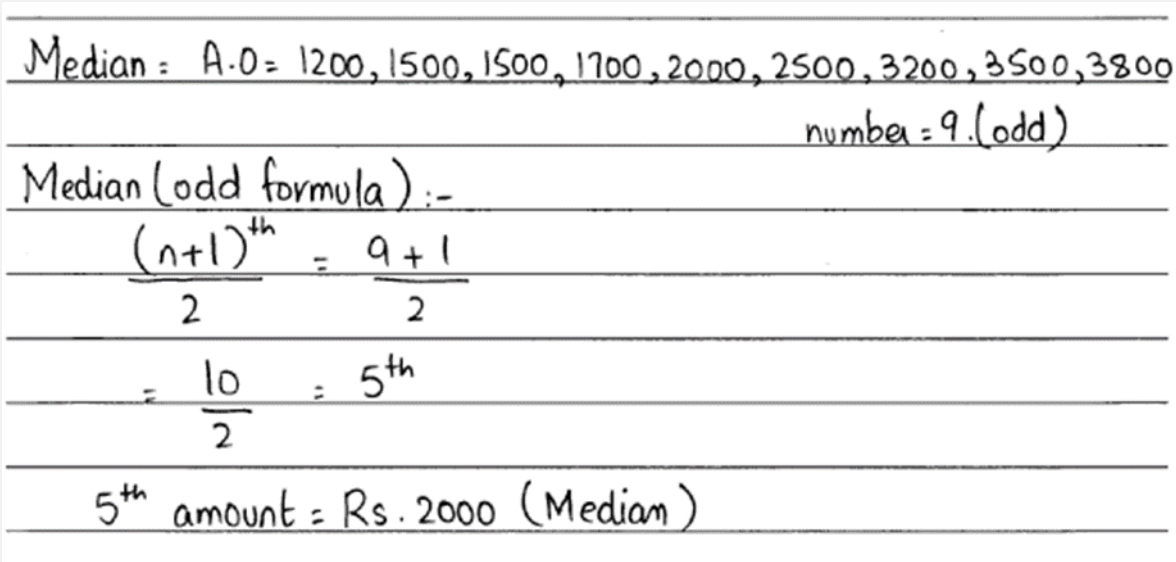
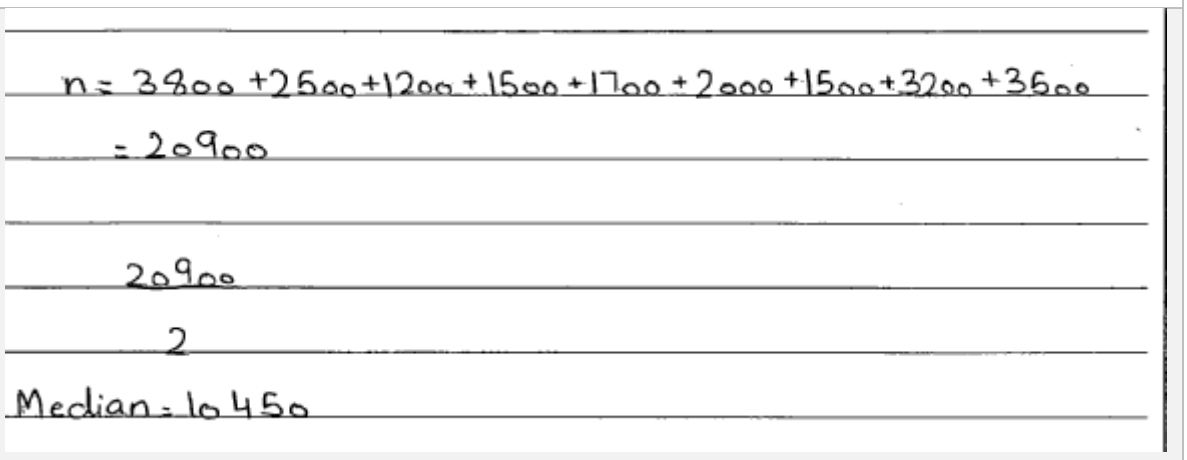
Images of Weaker Responses	 If the income tax rate is 12.5%, then calculate the i. monthly income tax. (1 Mark) 110000 × 12.5 / 100 13750 ii. monthly consumable income. (1 Mark) 15600 + 15000 + 10000 40000 + 13750 = 53750 consumable income iii. gross annual income. (1 Mark) 110 000 − 53750 = 56250 income left.

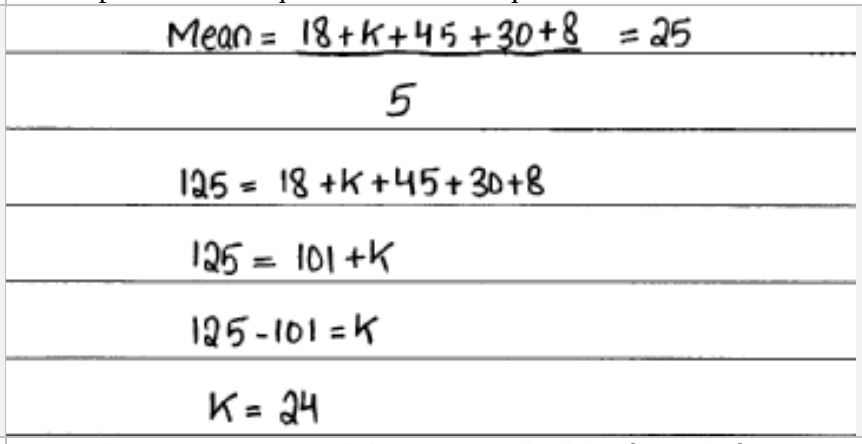
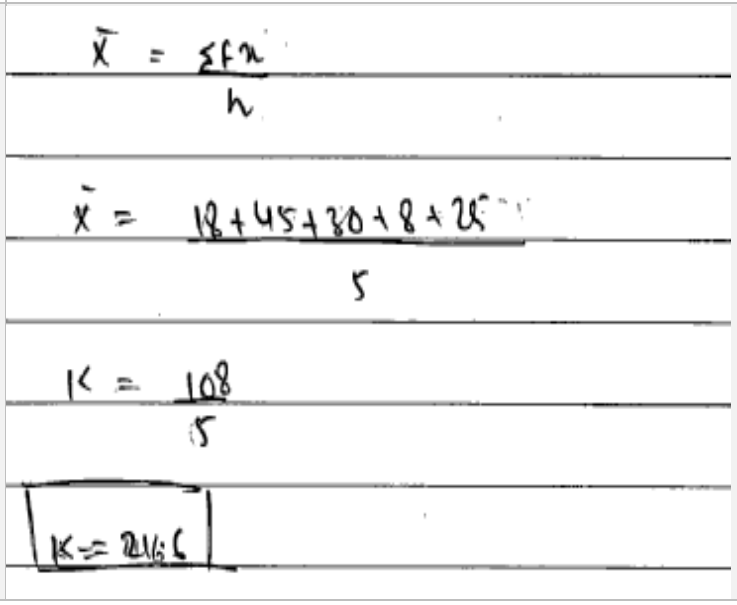
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:
Teachers should reinforce the conceptual understanding of income-related terms such as *gross income*, *tax*, and *net/ consumable income* through real-life examples and scaffolded practice. Introduce word problems that differentiate between monthly and annual figures to improve contextual comprehension. Regular formative assessments with feedback can help identify and address persistent misconceptions.

Question No. 2	
Question Text	i. On a certain day, 9 friends spend the following amount (in Rs) on a shopping trip. Find the median amount that was spent. 3,800, 2,500, 1,200, 1,500, 1,700, 2,000, 1,500, 3,200, 3,500
SLO No.	13.3.1

SLO Text	Calculate arithmetic mean, median and mode for ungrouped and grouped data.
Max Marks	2
Cognitive Level	A
Checking Hints	i. 1 mark for arranging in ascending order 1 mark for finding median
Overall Performance	Overall, candidates demonstrated a fair understanding of the concept of median and were largely successful in identifying its position. However, some struggled to arrange values in ascending order, which led to incorrect answers. A few confused the concept with mean or mode.
Description of Better Responses	In *better responses*, candidates correctly arranged the data in ascending order like 1,200, 1,500, 1,500, 1,700, 2,000, 2,500, 3,200, 3,500, 3,800, identified the number of values (9), and accurately applied the median formula (n+1)/2 to find the required position and value.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates incorrectly calculated the median by summing all values and dividing by 2, or by writing the mode instead of the median.
Image of Weaker Response	

Question No. 2	
Question Text	ii. If the mean of 5 numbers 18, *k*, 45, 30 and 8 is 25, then find the value of *k*.
SLO No.	13.3.1
SLO Text	Calculate arithmetic mean, median and mode for ungrouped and grouped data.
Max Marks	2
Cognitive Level	A
Checking Hints	ii. 1 mark for putting the values in equation of mean for ungrouped data 1 mark for finding value of k
Overall Performance	A few confused the concept with mean or mode. While most candidates correctly applied the mean formula $\overline{X} = \frac{\sum x}{n}$, common errors were observed in calculating the value of *k*, indicating a gap in comprehension of the underlying concept required by the question.
Description of Better Responses	They also demonstrated a clear understanding of calculating the mean for ungrouped data by substituting correct values and manipulating the equation $25 = \frac{18+k+45+30+8}{5}$ effectively to find the value of *k*. These responses reflected a strong grasp of both statistical concepts and the requirements of the question.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates misapplied the formula for grouped data despite the absence of frequencies, while others assumed *k* to be the mean $k = \frac{18+45+25+30+8}{5}$ without proper justification. These errors reflected a lack of conceptual clarity and led to incorrect answers.
Image of Weaker Response	

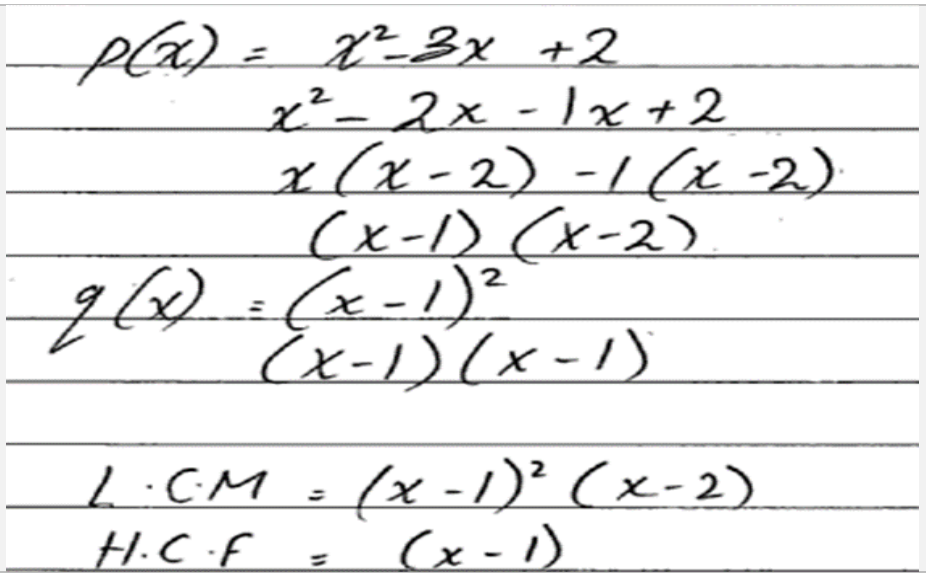
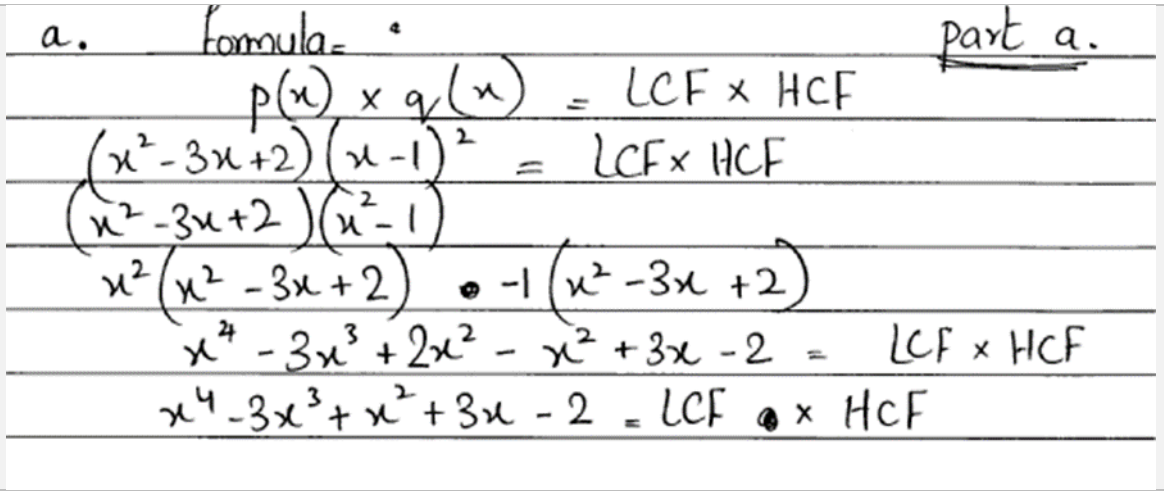
Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:		

Question No. 3a

Candidates were given the choice to attempt any ONE out of the two questions: 3a and 3b. Majority of the candidates attempted question 3b.

Question Text	It is given that $p(x) = x^2 - 3x + 2$ and $q(x) = (x-1)^2$ are two polynomials. For the given polynomials, find the LCM and HCF.
SLO No.	14.1.1
SLO Text	Find the highest common factor (HCF) and the least common multiple (LCM) of algebraic expressions using factorisation method.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for factorising $p(x) = x^2 - 3x + 2$ 1 mark for factorising $q(x) = (x-1)^2$ 1 mark for finding HCF. 1 mark for multiplying the remaining factors to find LCM.
Overall Performance	The responses indicated that good number of candidates demonstrated a sound conceptual understanding of factors, LCM, and HCF. However, some struggled with identifying correct factors and applying the appropriate methods, which led to incorrect attempts and outcomes. This suggests a need for reinforced teaching on foundational number properties and their applications in problem-solving contexts.
Description of Better Responses	In *better responses*, the candidates made correct factor of both the expressions by middle term breaking method $p(x) = x^2 - 3x + 2 = x^2 - 2x - x + 2$ and square expansion in two factor form. Furthermore, they exhibited a clear conceptual understanding of HCF, correctly identifying it as the single common factor. They then proceeded to multiply both the common and uncommon factors to express the LCM accurately, ultimately arriving at the correct result. This reflected both procedural fluency and a solid grasp of fundamental algebraic and number theory concepts.

Image of Better Response	
Description of Weaker Responses	*Weaker response*s showed that the candidates factorised the expression incorrectly $(x^2-3x+2)(x-1)^2$ and expanded it in the following step, which reflected a conceptual lack in understanding factorisation techniques. Furthermore, the candidates failed to demonstrate the correct concept of LCM and HCF. The product of the expressions was mistakenly written as the product of LCM and HCF, indicating confusion about the relationship between the two.
Image of Weaker Response	

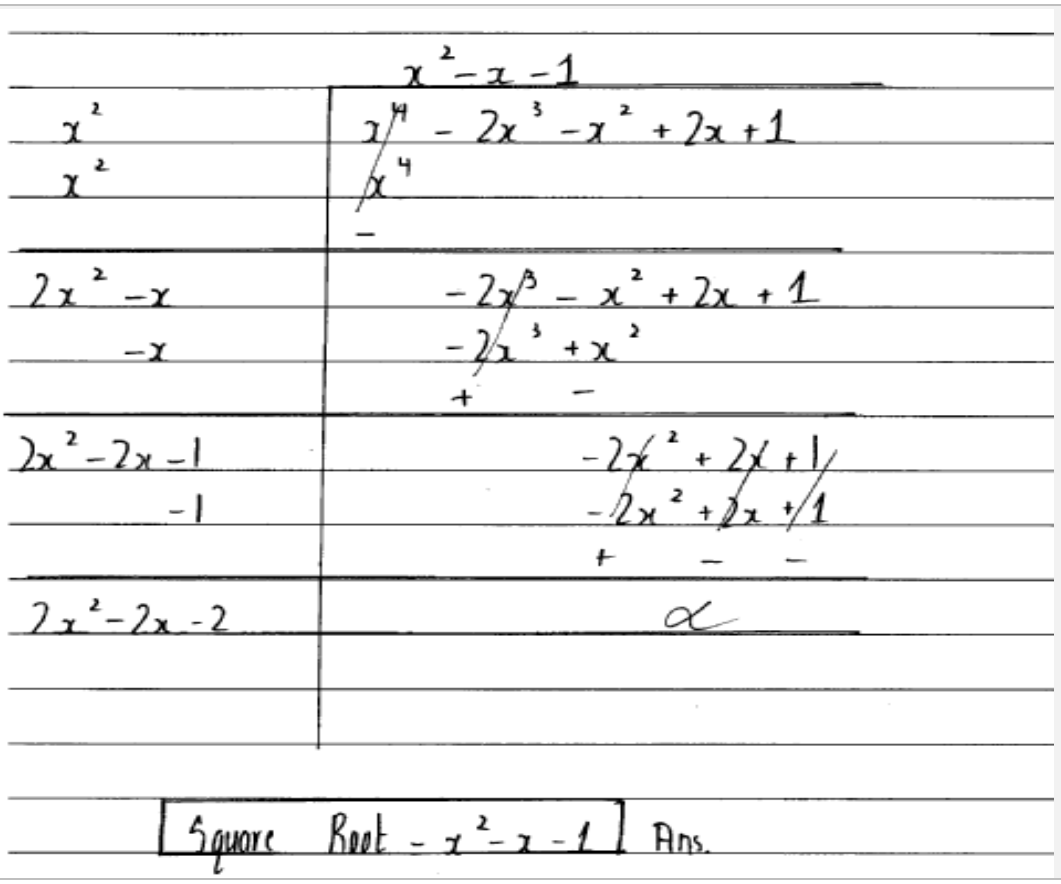
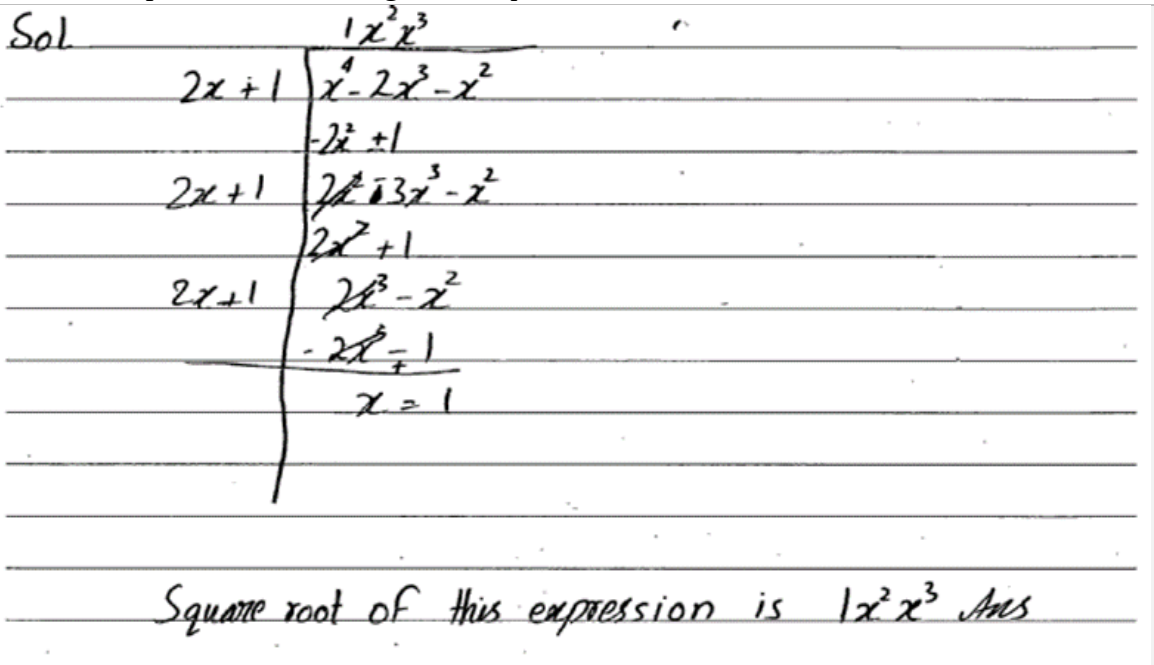
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies

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Any Additional Suggestion:
To strengthen students' understanding of LCM and HCF, teachers should emphasise the conceptual relationship between the two and reinforce the correct procedure for factorisation. Use targeted practice exercises that involve both middle term breaking and algebraic identities to develop fluency in factoring expressions. Visual aids such as factor trees or Venn diagrams can also help students distinguish between common and uncommon factors. Furthermore, teach the correct interpretation of the product of two expressions in relation to their LCM and HCF to avoid common misconceptions.

Question No. 3b	
Question Text	Find the square root of the expression $x^4 - 2x^3 - x^2 + 2x + 1$ by division method.
SLO No.	14.3.1
SLO Text	Calculate the square root of algebraic expression by factorisation and division.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for each steps of division (3 required) 1 mark for identifying the square root.
Overall Performance	A few candidates successfully completed all steps, demonstrating a clear understanding of the concept of square roots and applying the correct procedure. However, many struggled with the underlying concept required to approach the question appropriately.
Description of Better Responses	In *better responses*, candidates demonstrated a clear understanding of each step involved in the square root process. They correctly subtracted values at every stage and consistently followed the appropriate steps without any sign errors. The solution was carried through logically until the final step where ($-2x^2+2x+1$) cancelled out and the remainder was zero. Furthermore, the inclusion of the $\mp$ sign in the answer reflected their accurate conceptual understanding of square roots and their possible values.

Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates began with an incorrect multiplication e.g., (2x+1) with x^2 of the initial value, which affected all subsequent steps. Throughout the process, sign errors were evident, and similar values were incorrectly cancelled out. These mistakes indicated a weak conceptual understanding of the square root method.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

To address the conceptual gaps observed in weaker responses, it is recommended that teachers reinforce the step-by-step procedure of the square root method through targeted practice. Emphasis should be placed on the correct use of signs, understanding when and how to subtract values, and recognising the significance of obtaining a zero remainder. Incorporating visual aids such as create bingo cards with numbers; call out a square (e.g., "nine") and have students cover its square root (e.g., "81") on their cards and guided examples can help students build fluency and accuracy in such algebraic methods.

Question No. 4

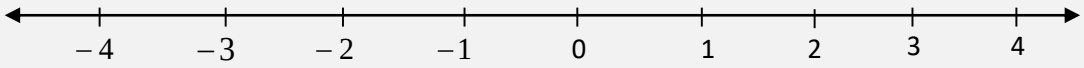
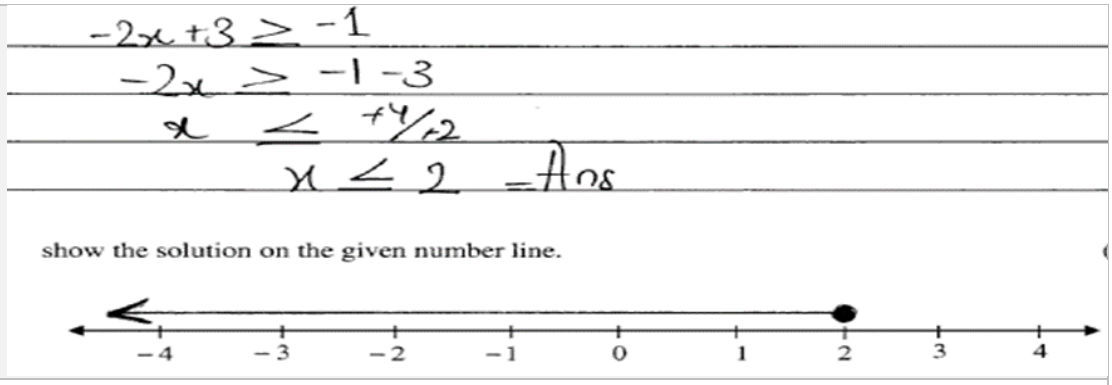
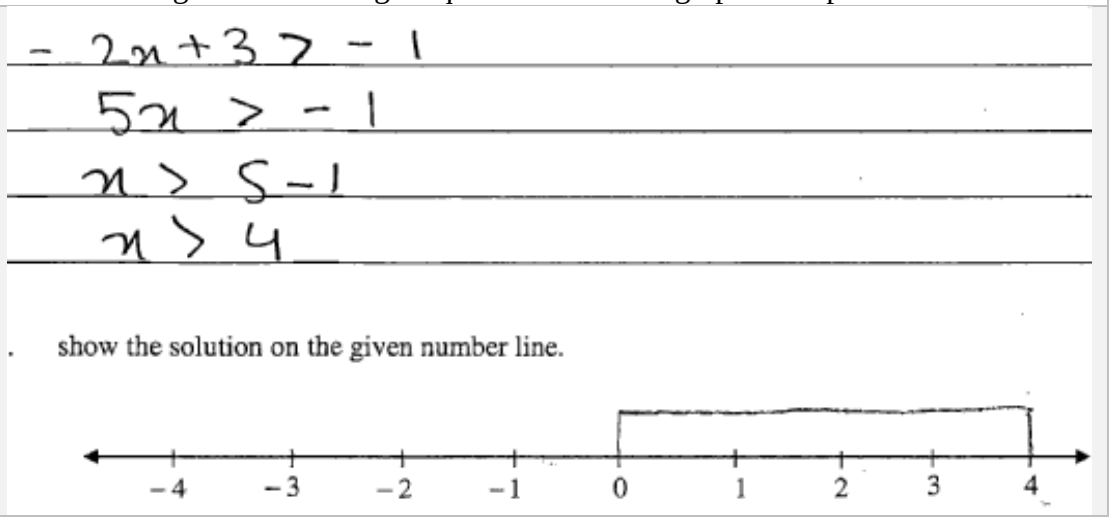
Question Text	If $x \in R$, then for $-2x+3 \geq -1$, i. solve for x. ii. show the solution on the given number line. 
SLO No.	15.4.1 & 15.4.2
SLO Text	Solve linear inequalities with real coefficients in one variable. Represent the solution of linear inequalities on the number line.
Max Marks	3
Cognitive Level	A
Checking Hints	i. 1 mark for subtracting 3 on both sides 1 mark for dividing by -2 on both sides ii. 1 mark for showing the solution on the number line
Overall Performance	This question aims to assess the concept of inequalities in open sentences and their representation on the number line. It was observed that few candidates struggled with handling inequalities in algebraic expressions, leading to errors in both their solutions and the solution representations on the number line.
Description of Better Responses	In *better responses*, the candidates correctly applied the operations of addition and subtraction within the context of inequalities. The accurate reversal of the inequality sign ($\geq$) when shifting terms across the equals sign demonstrated a sound conceptual understanding of the properties of inequalities. Furthermore, the number line was accurately drawn, clearly representing the solution set and showing a good grasp of the concept that x belongs to the set of real numbers.

Image of Better Response	
Description of Weaker Responses	In *weaker responses*, the candidates demonstrated an incorrect application of inequality rules by replacing –2 with an inappropriate mathematical operation. Furthermore, the number line was either missing or inaccurately drawn, and it failed to represent the value of *x* calculated by the candidates. This reflected a weak understanding of both solving inequalities and their graphical representation.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:
Teachers should provide targeted practice on the properties of inequalities, especially focusing on how the inequality sign changes when operations involve negative numbers. It is also recommended to reinforce the concept of representing inequalities on a number line, using varied examples. Interactive activities, such as

peer teaching or number line games, can help strengthen students' conceptual understanding and visual interpretation skills.

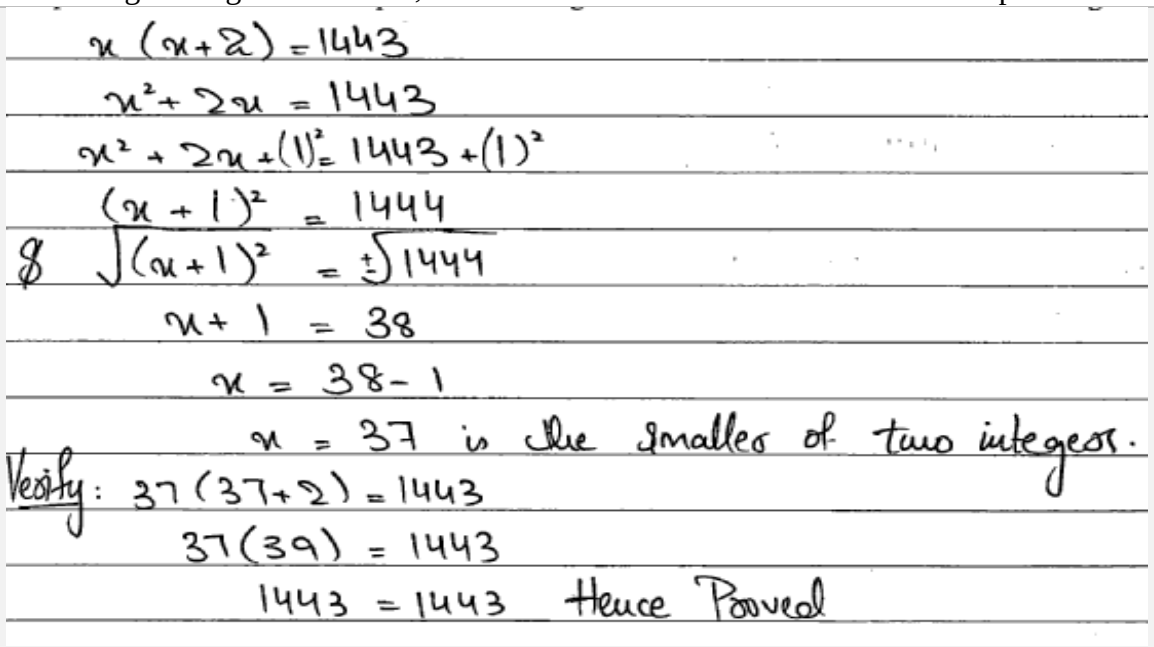
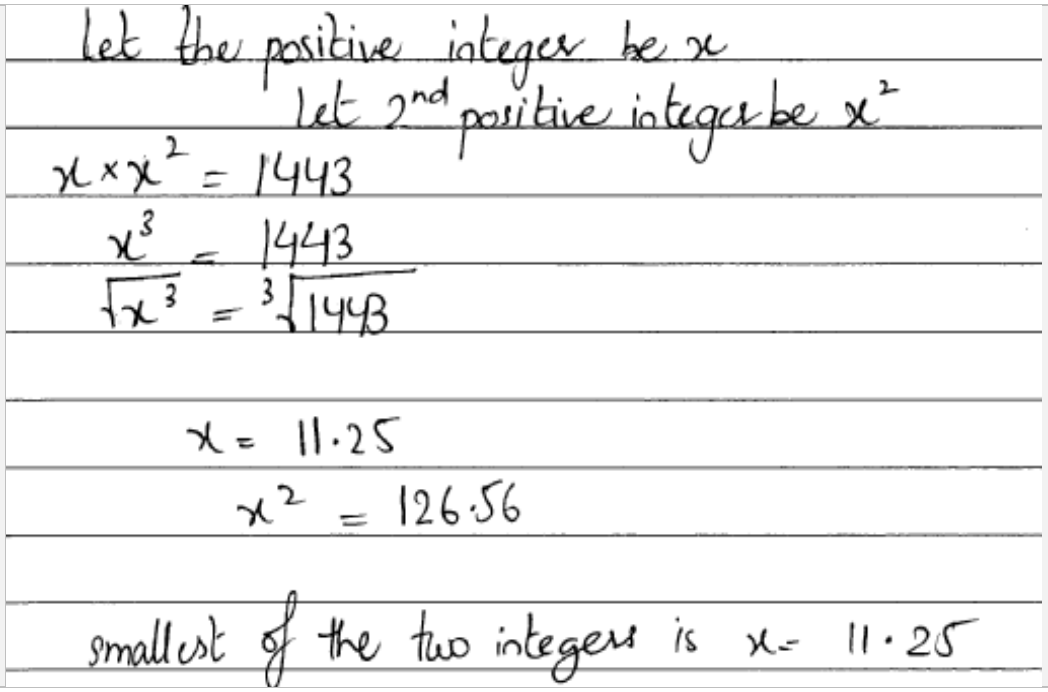
Question No. 5	
Question Text	The product of two consecutive positive odd integers is 1443. Find the two positive integers.
SLO No.	16.3.3
SLO Text	Solve word problems based on the quadratic equation.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for writing $x \times (x+2) = 1443$ 1 mark for complete factorisation 1 mark for finding the values of *x* 1 mark for explicitly mentioning the two positive integers
Overall Performance	In overall responses, a few candidates demonstrated excellent skills in comprehending the word problem and successfully proceeded to find the correct values of the two odd integers. However, the majority of candidates faced difficulty in formulating the correct algebraic statement, which was the main cause of losing marks in this question. The ability to interpret language and translate it accurately into mathematical notation appeared to be a significant challenge for many candidates.
Description of Better Responses	In *better responses*, candidates demonstrated sound understanding in comprehending the word problem. They correctly formulated the equation in quadratic form $x \times (x+2) = 1443$ and proceeded to solve it using complete square method $x^2 + 2x + (1)^2 = 1443 + (1)^2$. Candidates identified both values and showed strong conceptual understanding by discarding the negative solution, which was not relevant in the context of the problem.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates were unable to comprehend the given word problem effectively. They struggled with the concept of consecutive positive odd integers and, as a result, formed incorrect equation $x \times x^2 = 1443$. Some candidates attempted to find the value of *x* using a cube root ($\sqrt[3]{1443}$), which was not relevant or appropriate for the context of the question. This demonstrated a lack of understanding of the problem's requirements, ultimately leading to incorrect solutions.

Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

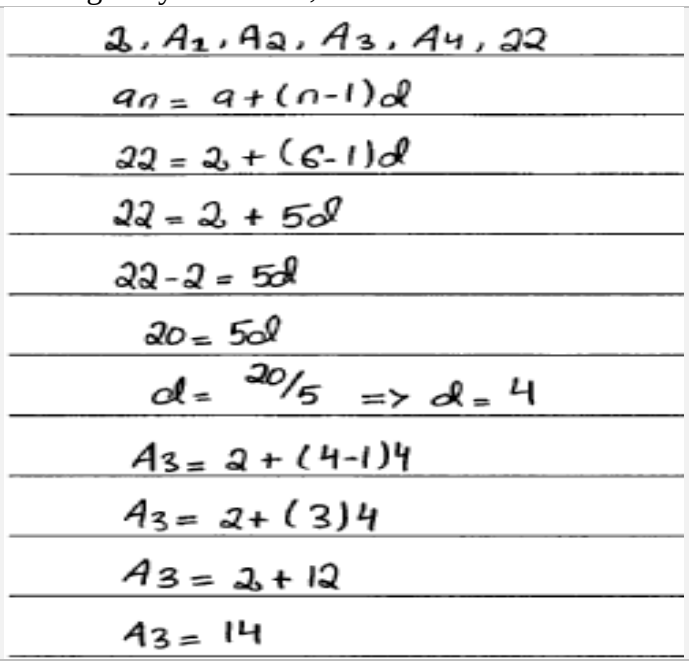
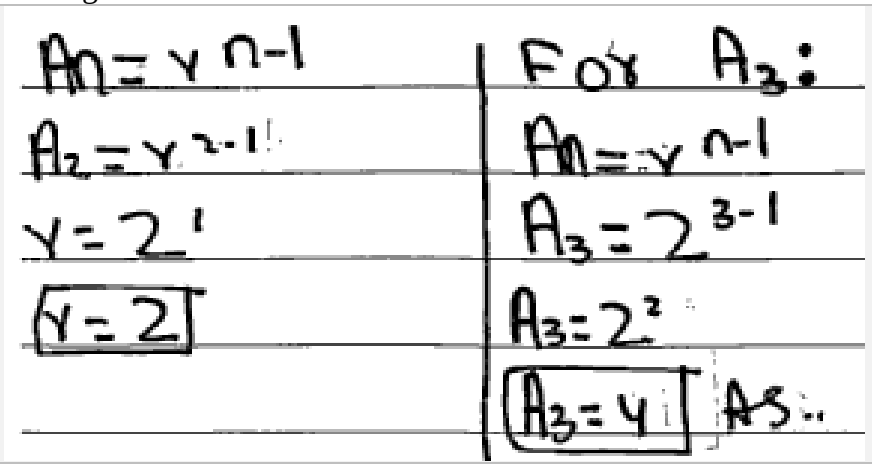
Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

- Integrate regular practice of translating word problems into algebraic expressions and equations during lessons.
- Include varied examples in class that focus on mathematical language interpretation.
- Encourage students to always validate their solutions by substituting the values back into the original scenario.

Question No. 6

Question Text	If A_1, A_2, A_3, and A_4 are arithmetic means between 2 and 22, then find the value of A_3.
SLO No.	17.3.2
SLO Text	Find n arithmetic means between two numbers, where n is a natural number.
Max Marks	4

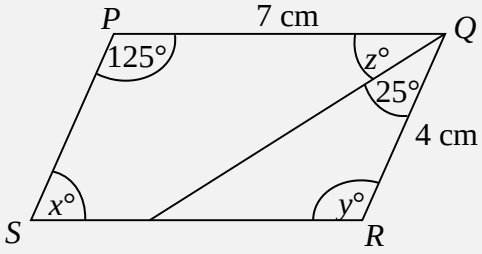
Cognitive Level	A
Checking Hints	1 mark for writing the means and given numbers in form of arithmetic sequence 1 mark for identifying the terms as $a_4 = A_3;\ a_6 = 22$ 1 mark for calculation of the value of d 1 mark for calculation of $A_3 = 2 + 3(4)$
Overall Performance	The overall performance in this question indicated that many candidates were able to comprehend the requirements. They demonstrated an understanding of how to represent the given terms and means in the form of an arithmetic sequence. The majority of candidates correctly identified the pattern and proceeded to determine the common difference, *d*, accurately.
Description of Better Responses	In *better responses*, candidates demonstrated a clear understanding of arithmetic sequences. They correctly applied the equation $22 = 2 + 5d$ to determine the value of the common difference (*d*). The concept was well grasped, and they proceeded to substitute the correct values into the appropriate equation $A_3 = 2 + 3(4)$ to find the required term. Their working was logically structured, which led them to successfully calculate the value of A_3.
Images of Better Responses	
Description of Weaker Responses	In *weaker responses*, candidates failed to distinguish between arithmetic mean (AM) and geometric mean (GM) and used formulae $A_n = r^{n-1}$ and (a+b)/2, which was not the appropriate method for this question. This reflected a misconception of the concept of arithmetic mean. As a result, they substituted incorrect values and followed an invalid process, ultimately leading to inaccurate answers.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:		

Question No. 7a

Candidates were given the choice to attempt any ONE out of the two questions: 7a and 7b. Majority of the candidates attempted 7a.

Question Text	In the given diagram, the quadrilateral *PQRS* is a parallelogram.  For the given parallelogram, find the i. angle *y*. ii. angle *x*. iii. angle *z*. iv. length of *SR*.
SLO No.	18.2.2
SLO Text	Apply the following properties of a parallelogram to find the unknown measurements a) opposite sides of a parallelogram are equal b) opposite angles of a parallelogram are equal.
Max Marks	4
Cognitive Level	U
Checking Hints	i. 1 mark for calculating angle *y* ii. 1 mark for calculating angle *x* iii. 1 mark for calculating angle *z* iv. 1 mark for finding length *SR*

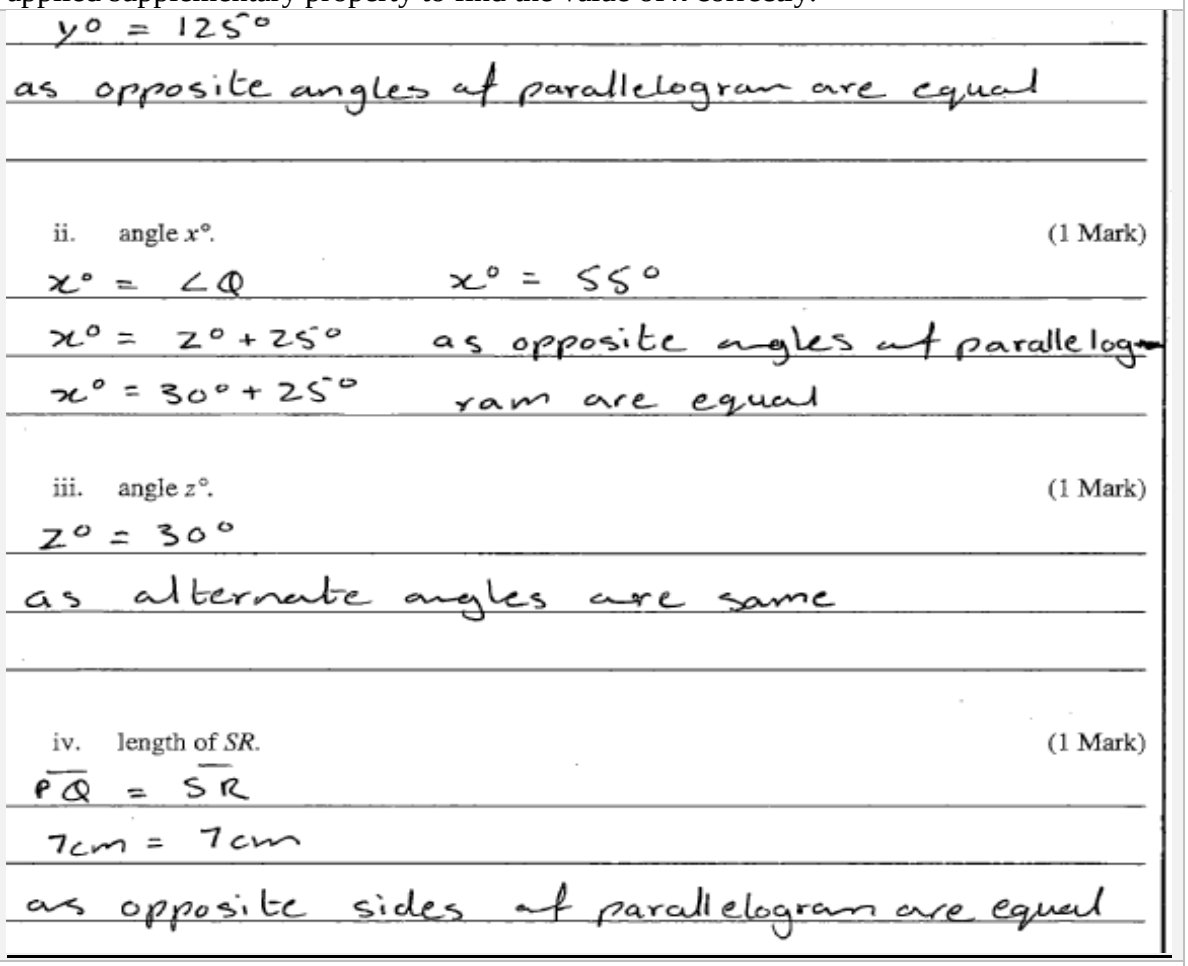
Overall Performance	Some candidates performed reasonably well in this question, however, only a few showed conceptual understanding regarding the properties of angles in a parallelogram. While many candidates correctly applied the properties related to the sides, they struggled to identify and calculate the required angles. The overall responses indicated difficulty in recalling or applying angle-related properties of parallelograms, which affected the accuracy of their answers.
Description of Better Responses	In *better responses*, candidates demonstrated a strong understanding of the properties of parallelograms. They accurately calculated the required angles and supported their answers by referencing relevant angle (opposite angles are equal) and side (opposite sides are equal and parallel) properties. Candidates correctly applied the concept of alternate angles. They applied supplementary property to find the value of *x* correctly.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* showed that candidates struggled to understand the fundamental properties related to the sides and angles of a parallelogram. They failed to write the correct required answers and incorrectly described angle *y* as a sum of 90°, indicating a conceptual gap in understanding the angle properties of a parallelogram. In part (iii), the equation formed lacked justification, and in several instances, candidates equated side lengths with angle measures, which led to inappropriate and mathematically invalid answers. At the end, when the question required finding the length of SR, the candidates wrote "90 degrees", indicating a lack of understanding of the distinction between measuring length and angle.

Image of Weaker Response	i. angle $y°$. (1 Mark) $25 + 25 = 190$ 50 Ans ii. angle $x°$. (1 Mark) $x = 125°$ / $125 + 125° = 90$ / $250 = 90$ $90 - 250$ $160°$ Ans iii. angle $z°$. (1 Mark) $25° + 125° = 180$ $150 = 180$ $= 30°$ Ans iv. length of SR. (1 Mark) $<xy$ semi circle $= 90°$

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

Teachers are encouraged to reinforce the conceptual difference between the measurement of angles (in degrees) and lengths (in units such as centimetres or metres) by measuring classroom objects for length and identifying angles within those objects. Incorporating real-life visual aids and geometry tools during

instruction can help students better grasp these distinctions. Moreover, providing targeted practice questions that require distinguishing between angle and length will support students in applying appropriate mathematical reasoning in context.

Question No. 7b

Candidates were given the choice to attempt any ONE out of the two questions: 7a and 7b.

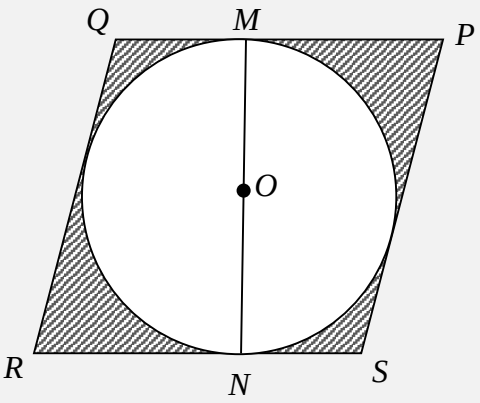
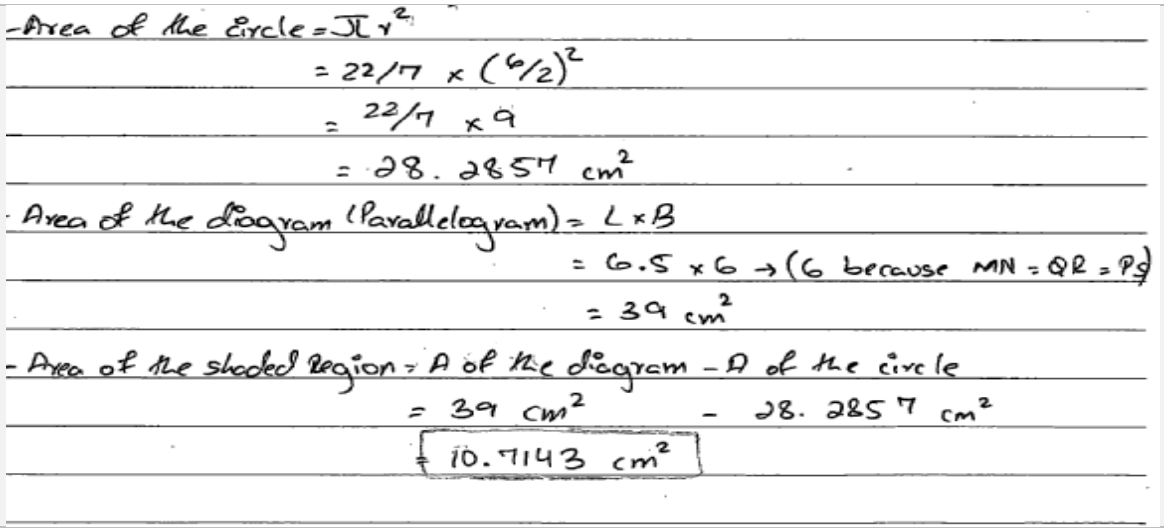
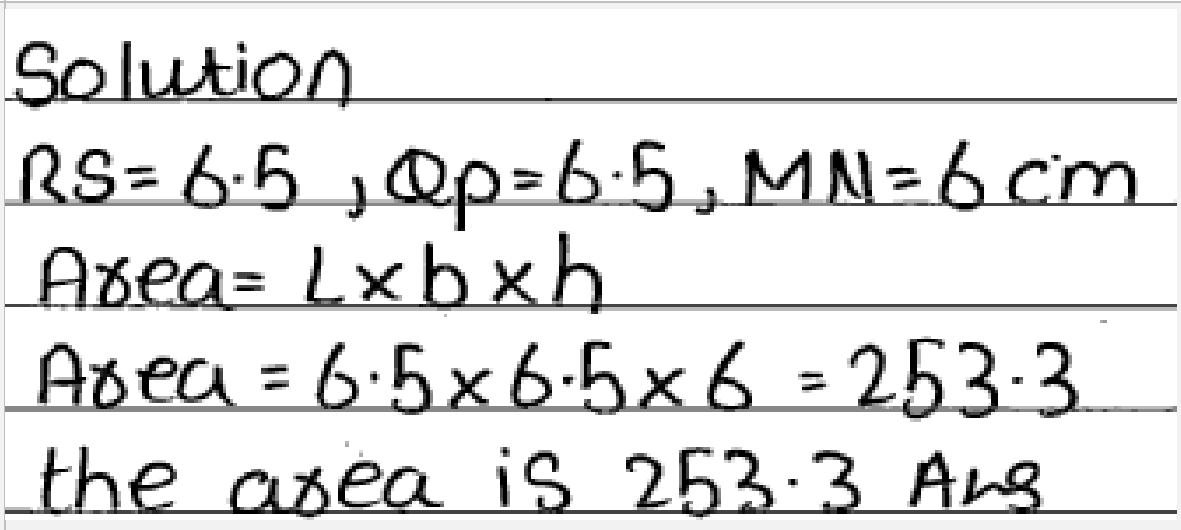
Question Text	In the given diagram, a circle with centre O is given. MN is the diameter of the circle and it is perpendicular to RS.  **NOT TO SCALE** If RS = 6.5 cm, QP = 6.5 cm and MN = 6 cm, then calculate the area of the shaded region.
SLO No.	19.2.4
SLO Text	Solve problems related with areas of triangle, rectangle, square, parallelogram and circle.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for calculating radius using diameter 1 mark for calculating the area of the circle 1 mark for calculating the area of the parallelogram 1 mark for calculating the shaded area
Overall Performance	Only a few candidates selected this question. These candidates applied the correct method and completed the question accurately. However, the majority of those who attempted the question struggled with identifying the region that needed to be shaded. They failed to grasp the fundamental concept behind the calculation, which led to incorrect working and ultimately an incorrect final answer.
Description of Better Responses	In *better responses*, candidates demonstrated a sound conceptual understanding of the areas of both a parallelogram and a circle. They took the square of radius $(3)^2 = 9$ and then multiplied with the value of π to find the area of circle. Furthermore, identified that the shaded region could be found by subtracting the area of the parallelogram from that of the circle. These candidates successfully applied this method and arrived at the correct numerical answer.

Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates were unable to grasp the underlying concept of the shaded region. Instead of applying area-based methods, they incorrectly proceeded with volume formulae (Volume of a cuboid $= l \times b \times h$), which reflected a fundamental misunderstanding of the question. Furthermore, the substitution of values into the formulae was also incorrect, indicating a lack of clarity in both concept and calculation. Overall, these responses showed weak conceptual understanding and misinterpretation of the task.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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	• Practical Demonstration	

Any Additional Suggestion:
To address the conceptual confusion between area and volume, teachers should incorporate more comparative problem-solving exercises that require students to distinguish between two-dimensional and three-dimensional contexts. Visual aids such as diagrams, physical models, and interactive geometry tools can also help reinforce the spatial understanding of shapes. Moreover, emphasising units (e.g., cm^2 for area vs. cm^3 for volume) during instruction and assessments can guide students to select the appropriate formulae for each context.

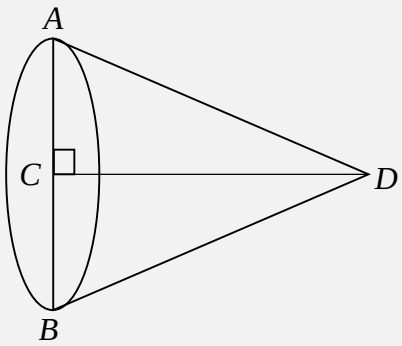
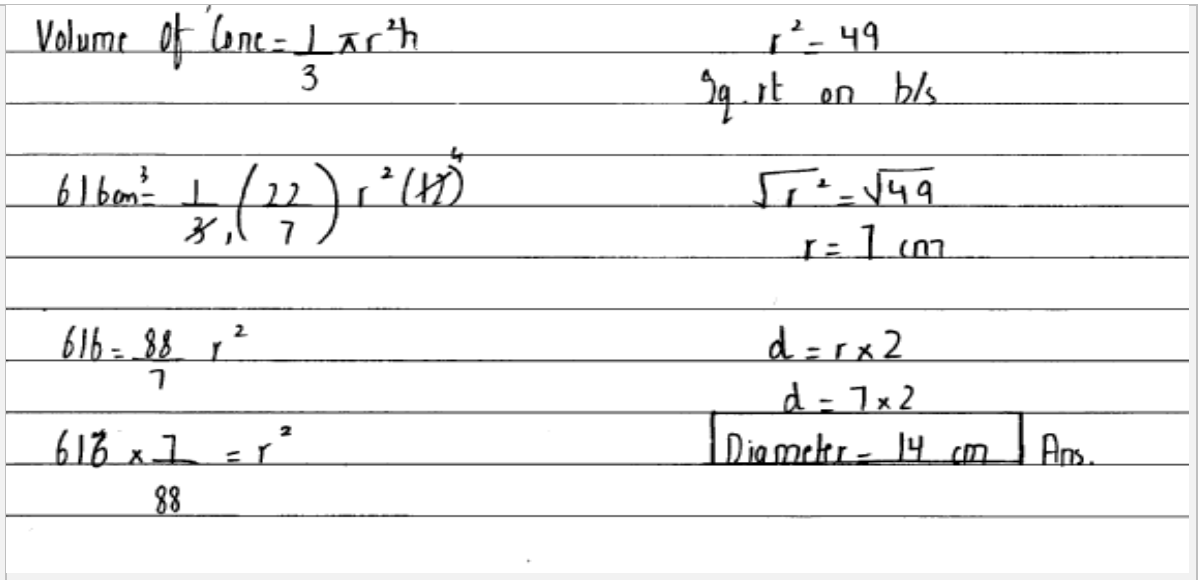
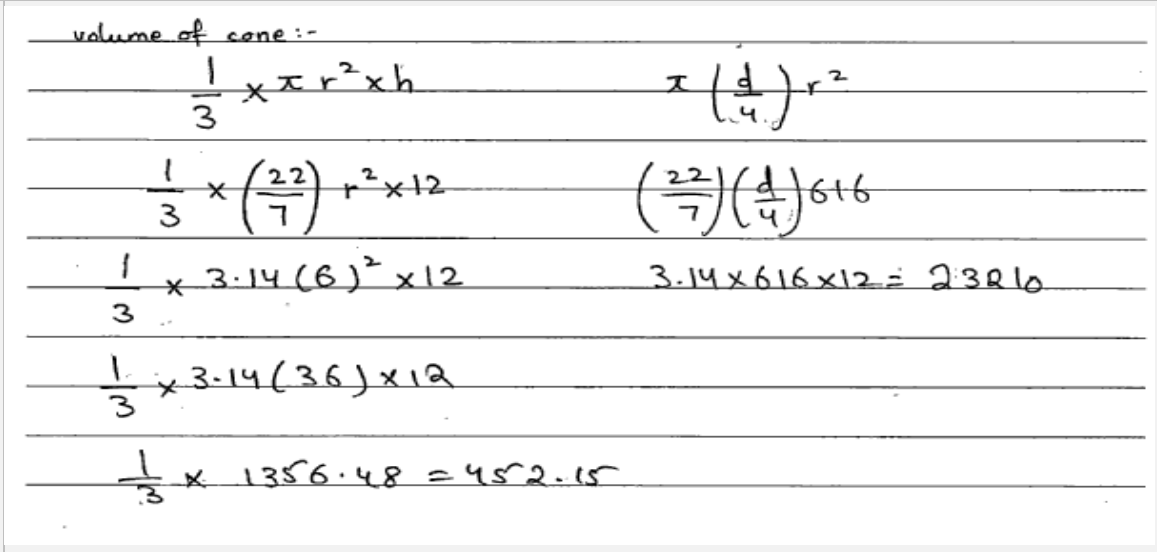
Question No. 8	
Question Text	The given diagram shows a right circular cone.  **NOT TO SCALE** Calculate the diameter of the cone if its height is 12 inches and volume is 616 cubic inches. (**Note**: Take $\pi = \frac{22}{7}$)
SLO No.	19.3.1d
SLO Text	Find the volume of d) a right circular cone when its radius and height are given.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for identifying the formula for volume of cone 1 mark for substituting the given values 1 mark for calculating the radius 1 mark for finding the diameter
Overall Performance	Overall, responses were good. Most candidates found the question straightforward, as the formula (Volume of circular cone = $\pi r^2 \times \frac{h}{3} \text{inch}^3$) was provided and they interpreted it correctly. However, some candidates failed to recognise the appropriate formula and made errors in substituting the correct values.
Description of Better Responses	In the *better response*, the candidates applied the correct values in the appropriate formula (Volume of circular cone = $\pi r^2 \times \frac{h}{3} \text{inch}^3$) and accurately calculated the radius as r = 7. The candidates cancelled out 3 and 12 with the table of 3 perfectly and demonstrated correct evaluation of the given values to determine the radius, and subsequently calculated the diameter by multiplying the radius by 2 to get it 14. This reflected a good understanding of the concept.

Image of Better Response	
Description of Weaker Responses	In *weaker responses*, the candidates correctly applied the formula but did the wrong substitution of $r = 6$, despite the fact that the radius was to be calculated. Additionally, the candidates failed to substitute the given volume value of 616 on the left-hand side of the equation, which led to an incorrect final answer. The radius calculated was therefore inaccurate, and the step for calculating the diameter was also omitted. This reflected a limited understanding of the procedure required.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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• Utilise the resource guide for additional materials	• Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion:

Teachers are encouraged to place greater emphasis on helping candidates understand the logical sequence of solving formula-based problems. Explicitly teaching the importance of correctly substituting known values, identifying what needs to be calculated, and interpreting given information (e.g., volume or area) can reduce careless errors. Incorporating step-by-step guided practice and encouraging students to verify each step can strengthen their procedural fluency and conceptual understanding.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing

with the entire class. It fosters active participation, communication skills, and diverse perspectives.

Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)

Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.

Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration

Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.

Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

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Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies along with suggestions for improvement.

These contributors include:

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